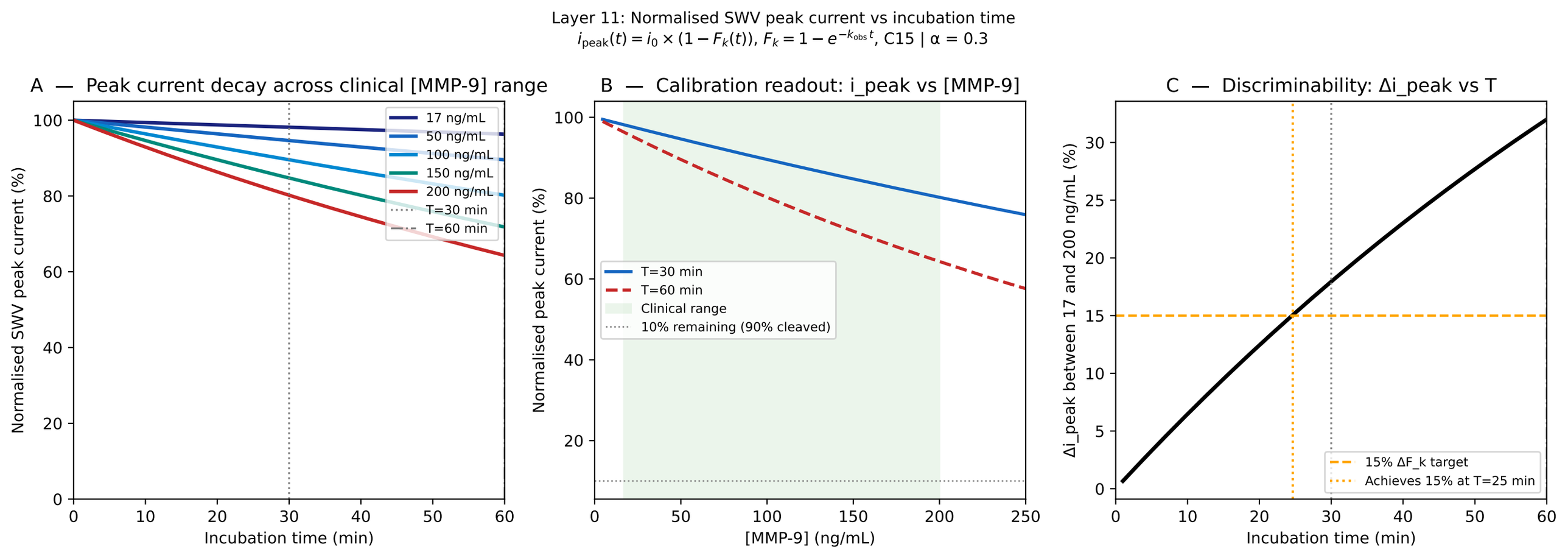
Layer 11: Normalised SWV peak current vs incubation time
$i_{peak}(t) = i_0 \times (1 - F_k(t))$, $F_k = 1 - e^{-k_{obs}t}$, C15 | α = 0.3
A — Peak current decay across clinical [MMP-9] range
Normalised SWV peak current (%)
Incubation time (min)
17 ng/mL
50 ng/mL
100 ng/mL
150 ng/mL
200 ng/mL
T=30 min
T=60 min
B — Calibration readout: i_peak vs [MMP-9]
Normalised peak current (%)
[MMP-9] (ng/mL)
T=30 min
T=60 min
Clinical range
10% remaining (90% cleaved)
C — Discriminability: Δi_peak vs T
Δi_peak between 17 and 200 ng/mL (%)
Incubation time (min)
15% ΔF_k target
Achieves 15% at T=25 min